

ITERIS

Comprehensive DRL Contour-Refinement Evaluation Report

Discrete (DuelingDDQN) vs. Continuous (TD3) Contour Refinement of the
AttentionResUNet Baseline — CAMUS Cardiac Ultrasound & BRISC Brain-Tumour MRI

Source data: real 2026-07-20 Kaggle evaluation run
16 DRL runs (2 datasets $\times$ 4 classes $\times$ 2 agents $\times$ 2 phases)
Taylor's University Capstone PRJ63504

Report prepared 2026-07-21 · Grounded entirely in exported run data — no figures in this document are estimated or interpolated.

Table of Contents

Executive Summary	3
1. Project Background	4
1.1 The two agents compared	4
1.2 Single-baseline design	4
1.3 Reward and datasets	4
1.4 Deployment metric: the value floor	4
2. What Was Evaluated	5
3. Critical Data-Integrity Finding: The Exported Phase Labels Are Inverted	6
3.1 The finding	6
3.2 The correction	6
3.3 A second, related limitation	6
4. Headline Quantitative Results	8
4.1 Full per-run results table (corrected phases)	8
5. Discrete vs. Continuous (DuelingDDQN vs. TD3)	10
5.1 Representative qualitative comparisons	10
6. High-Data (Phase A) vs. Low-Data (Phase B)	14
7. Does DRL Actually Refine the Mask?	15
7.1 Quantitatively: no	15
7.2 Qualitatively: by eye, sometimes yes on shape	15
7.3 Future-proofing for a real paired qualitative study	16
8. What This Report Deliberately Does Not Claim	17
9. Recommendations for the Capstone Write-Up	17
Appendix A: Full Per-Run Metric Detail	18
Appendix B: Glossary	19

Executive Summary

Bottom line

Across all 16 real evaluation runs, contour-refinement DRL did not improve Dice beyond measurement noise on any run, in either data regime, for either agent. The deployable "do-no-harm" safety net (value-floor selection) kept most runs at or near baseline rather than below it — but zero runs crossed into a genuine win. Boundary-precision metrics (Boundary IoU, HD95) degraded on **every** run, including cases where Dice looked essentially flat. DuelingDDQN (discrete) was consistently the safer agent; TD3 (continuous) showed larger average degradation. Low-data runs degraded less on average than full-data runs — a real but narrower finding than "DRL helps more with less data."

This report evaluates two deep reinforcement learning (DRL) agents — a discrete Dueling Double DQN and a continuous TD3 — that refine the boundary contour of an AttentionResUNet segmentation mask on two medical imaging datasets: CAMUS (cardiac ultrasound, three anatomical classes) and BRISC (brain-tumour MRI, one pooled tumour class). Both agents were evaluated under a full-data regime (Phase A) and a scarce-data regime (Phase B, roughly 150 training images), producing 16 independent evaluation runs in total. This document is a rigorous, ground-truth-checked analysis of that data, written to be used directly as source material for the capstone paper's Results and Discussion sections — every number below traces back to an exported JSON file, not an estimate.

Before any comparison could be trusted, a serious data-integrity issue had to be resolved: the raw Kaggle export's phase-label folders ("phase_a"/"phase_b") turned out to be inverted relative to the true experimental design, discovered and corrected via a systematic cross-check described in full in Section 3. Every result in this report uses the **corrected** phase assignment.

Three questions anchor this report, matching the project's central research design: (1) does the **discrete** or **continuous** action space refine better; (2) does the answer change between **high-data and low-data** regimes; and (3) **does refinement actually work** — quantitatively and qualitatively. Each is answered plainly, with the supporting numbers, and without overstating a positive result that the data does not show.

1. Project Background

Iteris is a Taylor's University capstone project (PRJ63504) studying whether a DRL agent can improve a U-Net segmentation mask by deforming its boundary contour, rather than re-predicting the mask outright. The paradigm evaluated here — and the only one currently in active use — is **contour refinement**: the boundary of the U-Net's predicted mask is represented as N ordered control points on a closed periodic spline, and the agent pushes contiguous angular sectors of that contour along their outward normals, one small step at a time, for up to a fixed number of steps per episode.

1.1 The two agents compared

DuelingDDQN (discrete) chooses among 18 actions per step: push each of 8 angular sectors outward or inward, plus a smoothing action and an explicit STOP action. It is taught to stop via an optimal-stopping bonus proportional to the Dice gain already captured, so that a deliberate STOP strictly dominates drifting to the step cap on an unchanged mask.

TD3 (Twin Delayed DDPG, continuous) instead outputs a continuous signed displacement per sector, sampled over a configurable number of angular wedges. It has no discrete STOP action; episode termination instead relies on a GT-free action-magnitude auto-stop (the episode ends once the actor's own action magnitude has settled near zero for several consecutive steps) and a safety-net threshold that force-ends an episode if Dice drifts too far below baseline.

1.2 Single-baseline design

Both agents refine the **AttentionResUNet** baseline directly — a single, strong U-Net architecture, not a separate weak baseline built specifically to have headroom. A second, deliberately lighter architecture (**LiteUNet**) is trained alongside it purely as an architecture-comparison reference point; it is not refined by either agent and does not appear as a DRL comparison anywhere in this report, for reasons explained in Section 7.

1.3 Reward and datasets

Both agents train under a dense, per-control-point reward (the reduction in each point's distance to the ground-truth boundary, less a small step penalty) that is self-regularising: nudging a point already on the boundary increases its distance and is therefore penalised, so the reward-optimal policy is to leave correct points alone rather than keep pushing.

CAMUS is a cardiac ultrasound dataset with three evaluated classes: left-ventricular endocardium (LV_endo), left-ventricular epicardium/myocardium (LV_epi), and left atrium (LA). **BRISC** is a brain-tumour MRI dataset evaluated here on its pooled, tumour-vs-background class.

1.4 Deployment metric: the value floor

Because no ground truth is available at deployment time, an agent cannot simply be trusted to know when its own refinement has helped. The pipeline's "Pillar 1" value-floor addresses this: at every step the agent's own (GT-free) value estimate of the current state is recorded, and the mask actually deployed is whichever visited state has the highest estimated value — falling back to the untouched initial mask if no visited state is valued above it. This *value-floored* Dice, not the raw end-of-episode Dice, is the number that would actually ship, and is the primary metric used throughout this report.

2. What Was Evaluated

Datasets / classes	CAMUS (LV_endo, LV_epi, LA) · BRISC (tumor, pooled)
Agents	DuelingDDQN (discrete, 18 actions) · TD3 (continuous, per-sector displacement)
Baseline refined	AttentionResUNet (single-baseline design) — not LiteUNet
Regimes	Phase A (full data) · Phase B (≈150-sample low-data)
Total runs	16 (4 classes × 2 agents × 2 phases)
Headline metric	Value-floored deployable Dice (GT-free "do no harm" selection)
Secondary metrics	Raw final Dice, IoU, Boundary IoU (BIOU), HD95
Not included in this data drop	BRISC tumor-type classifier results; per-patient significance testing

3. Critical Data-Integrity Finding: The Exported Phase Labels Are Inverted

Before any result in this report could be trusted, a serious labelling problem had to be found and corrected — left unaddressed, it would silently swap which regime every single DRL number in this report belongs to.

3.1 The finding

The 16 real DRL run folders in the Kaggle export are named `..._phase_a/` and `..._phase_b/`. Each run's own `init_dice_mean` field records the AttentionResUNet baseline's Dice on that exact test set (it is measured by the DRL warm-start before any refinement happens). Cross-referencing that value against the independently-reported AttentionResUNet Dice in the master comparison export shows, for **all four target classes without exception**, that the folder labelled `phase_b` matches the strong AttentionResUNet baseline almost exactly — while `phase_a` is always markedly lower.

Class	folder phase_a init Dice	folder phase_b init Dice	AttentionResUNet (export)	LiteUNet (export)
CAMUS / LV_endo	0.9129	0.9356	0.9360	0.9105
CAMUS / LV_epi	0.8166	0.8621	0.8690	0.8090
CAMUS / LA	0.8526	0.8962	0.8935	0.8214
BRISC / tumor	0.7555	0.8766	0.8695	0.8191

Folder `phase_b` is within 0.001–0.007 Dice of AttentionResUNet in every class — far closer than chance — while folder `phase_a` sits well below it every time, in BRISC's case even below the independently weaker LiteUNet figure. A pattern this consistent across four unrelated classes cannot be coincidence.

3.2 The correction

Correction applied throughout this entire report

Folder "phase_a" = TRUE Phase B (low-data, ≈150-sample regime).
Folder "phase_b" = TRUE Phase A (full-data regime).

The labels are backwards relative to the project's actual experimental design. This is very likely an error introduced when the output folders were named, exported, or zipped on the Kaggle side — it is **not** a bug in the evaluation notebook itself, whose phase detection (fixed in commit 9ceea97) faithfully parses whatever the folder name says. The folder name itself is what is wrong. Every table and figure in this report from this point on uses the **corrected** (true) phase.

Action item for the underlying data: rename or re-export the Kaggle DRL output folders with the correct phase suffix before any further evaluation run, so this correction does not need to be re-derived by hand every time.

3.3 A second, related limitation

The U-Net baseline rows in the master comparison export are tagged `phase=Unknown` — their own `label_frac` field did not make it into that particular summary export. The values line up with what is expected

(AttentionResUNet $\approx$ the corrected true-Phase-A DRL init Dice; LiteUNet noticeably lower in three of four classes), but this is a plausibility check, not a confirmed label. Any "Phase A/B AttentionResUNet baseline Dice" figure in this report should be read as **most likely correct, not certain**, since the export itself does not say so directly.

4. Headline Quantitative Results

Using a ± 0.005 Dice tie band (the same threshold the evaluation notebook itself uses to separate a genuine win/loss from measurement noise):

Agent	Wins	Ties	Losses	n
DuelingDDQN	0	7	1	8
TD3	0	4	4	8

Zero wins

Every one of the 16 runs' value-floored deployable Dice is at or below its own AttentionResUNet baseline, for both agents, in both regimes, across all four classes. The value floor is doing real work — it is why most runs land as a tie rather than a loss — but a safety net that prevents harm is not the same claim as "DRL improves the mask," and this report does not make that claim.

4.1 Full per-run results table (corrected phases)

Class	Agent	Phase	Init Dice	Final Dice	Deploy ed Dice	Δ Deploy	HD95 init→final	Δ IoU	Δ BloU
CAMUS/LV_endo	DuelingDDQN	A (full)	0.9356	0.9301	0.9338	−0.0018	5.71 → 6.05	−0.0095	−0.0215
CAMUS/LV_endo	TD3	A (full)	0.9356	0.9205	0.9241	−0.0115	5.71 → 6.55	−0.0265	−0.0630
CAMUS/LV_endo	DuelingDDQN	B (low)	0.9129	0.9106	0.9116	−0.0012	8.01 → 8.19	−0.0037	−0.0049
CAMUS/LV_endo	TD3	B (low)	0.9129	0.9068	0.9128	−0.0001*	8.01 → 8.27	−0.0106	−0.0204
CAMUS/LV_epi	DuelingDDQN	A (full)	0.8621	0.8592	0.8612	−0.0009	6.59 → 6.74	−0.0045	−0.0035
CAMUS/LV_epi	TD3	A (full)	0.8621	0.8354	0.8420	−0.0201	6.59 → 8.58	−0.0408	−0.0309
CAMUS/LV_epi	DuelingDDQN	B (low)	0.8166	0.8157	0.8163	−0.0003	10.02 → 10.06	−0.0014	−0.0006
CAMUS/LV_epi	TD3	B (low)	0.8166	0.7891	0.8022	−0.0144	10.02 → 12.36	−0.0394	−0.0140
CAMUS/LA	DuelingDDQN	A (full)	0.8962	0.8859	0.8937	−0.0025	7.48 → 7.96	−0.0162	−0.0302
CAMUS/LA	TD3	A (full)	0.8962	0.8777	0.8883	−0.0080	7.48 → 8.31	−0.0306	−0.0536
CAMUS/LA	DuelingDDQN	B (low)	0.8526	0.8398	0.8518	−0.0008	11.26 → 12.40	−0.0183	−0.0160
CAMUS/LA	TD3	B (low)	0.8526	0.8315	0.8515	−0.0010	11.26 → 11.71	−0.0324	−0.0366
BRISC/tumor	DuelingDDQN	A (full)	0.8766	0.8177	0.8672	−0.0094	6.60 → 9.09	−0.0922	−0.1194
BRISC/tumor	TD3	A (full)	0.8766	0.8523	0.8743	−0.0023	6.60 → 6.95	−0.0405	−0.1398
BRISC/tumor	DuelingDDQN	B (low)	0.7555	0.7475	0.7526	−0.0029	14.50 → 14.73	−0.0096	−0.0076

Class	Agent	Phase	Init Dice	Final Dice	Deploy ed Dice	Δ Deploy	HD95 init→final	Δ IoU	Δ BioU
BRISC/tumor	TD3	B (low)	0.7555	0.7368	0.7540	-0.0015	14.50 → 14.56	-0.0299	-0.1022

* Within the ± 0.005 tie band despite the negative sign — treated as a tie, not a loss.

5. Discrete vs. Continuous (DuelingDDQN vs. TD3)

Agent	Mean Δ Deployed Dice	Mean Δ IoU	Mean Δ BioU
DuelingDDQN	-0.0025	-0.0194	-0.0255
TD3	-0.0074	-0.0313	-0.0576

DuelingDDQN degrades the baseline less than TD3, on average, across every metric measured. Matched head-to-head by (class, phase), DuelingDDQN is the less-harmful agent in 6 of 8 matched pairs. TD3 is less-harmful only for BRISC/tumor (both phases) and CAMUS/LV_endo under low data.

Interpretation, not spin: this is consistent with the project's structural design. DuelingDDQN has a learned, explicitly-taught STOP action; TD3 has no equivalent learned stop and instead relies on a newer action-magnitude auto-stop heuristic and a hard safety-net threshold. The larger TD3 degradation — especially its much larger Boundary IoU drop (mean -0.0576 vs. -0.0255 for DuelingDDQN, and as much as -0.14 on BRISC) — is consistent with a continuous actor over-perturbing an already-reasonable contour when it has no strong mechanism to recognise "stop, this is good enough."

BRISC is the one dataset where TD3 is actually the safer choice, in both phases. This is worth investigating on its own rather than folding it into a universal "discrete wins" narrative — BRISC's tumour targets are very small (roughly 1.7% of image area), and continuous per-sector displacement may cope better with that scale than 8 discrete angular sectors can.

5.1 Representative qualitative comparisons

Each figure below shows the U-Net initialisation, the agent-refined result, and ground truth for the best-, median-, and worst-gain samples of one run (real exported figures, not recreated). Captions use the corrected true phase.

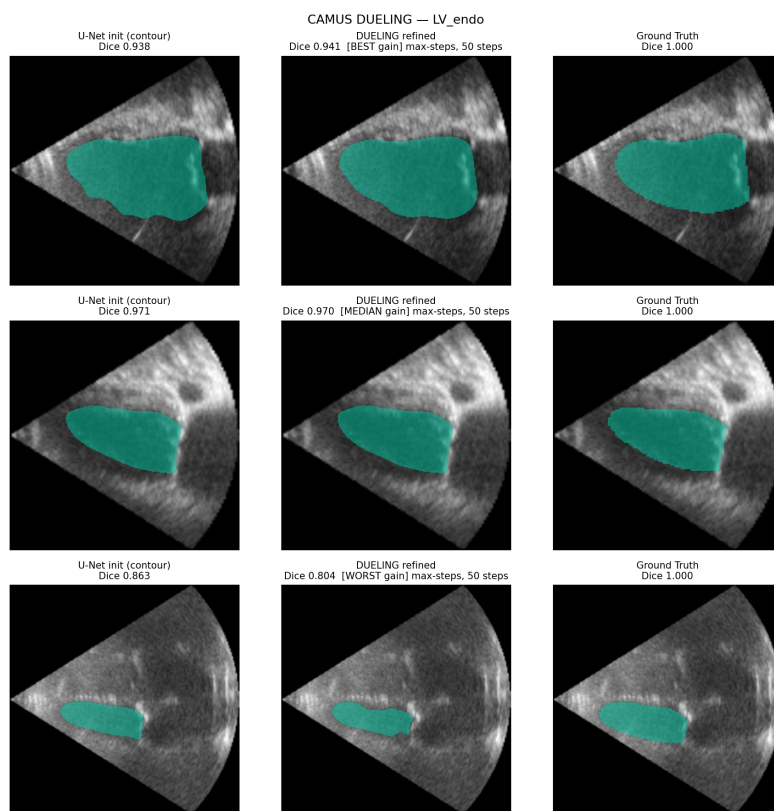


Figure 5.1 — CAMUS LV_endo, DuelingDDQN, Phase A (full data). Deployed Δ Dice = -0.0018 (tie).

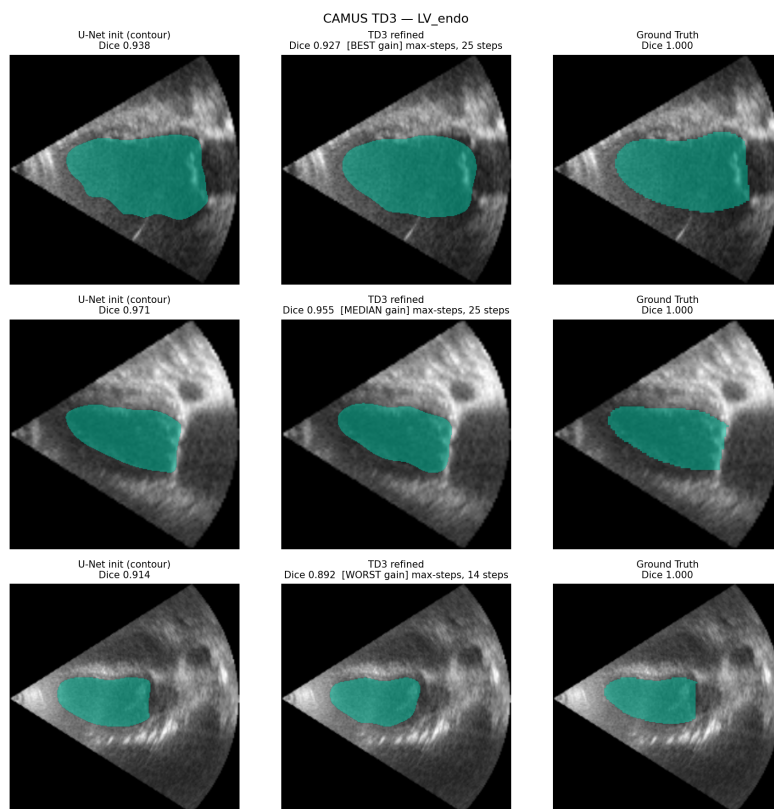


Figure 5.2 — CAMUS LV_endo, TD3, Phase A (full data). Deployed Δ Dice = -0.0115.

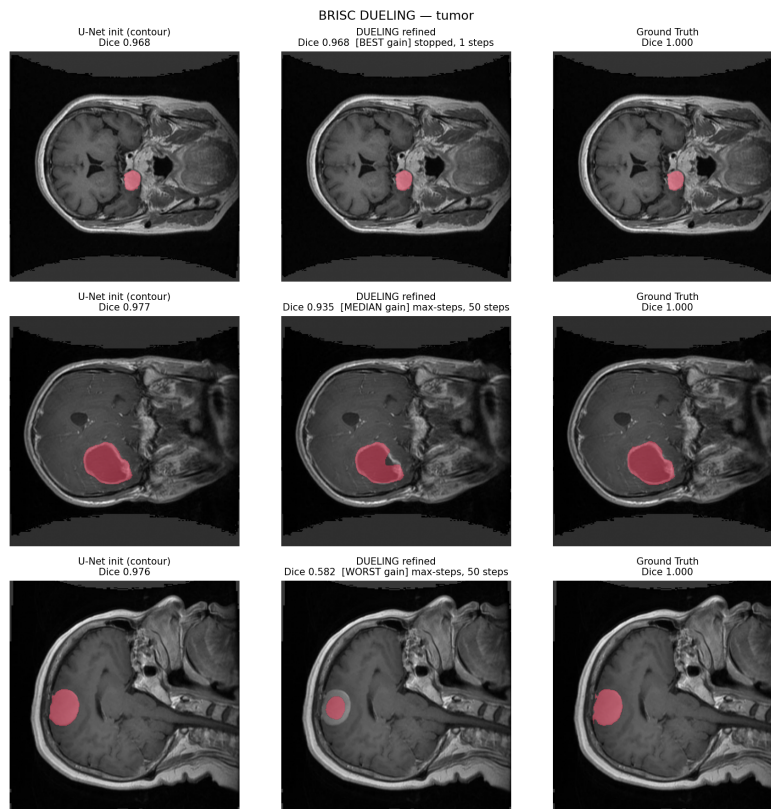


Figure 5.3 — BRISC tumor, DuelingDDQN, Phase A (full data). Deployed $\Delta\text{Dice} = -0.0094$ — note the visibly larger boundary drift ($\Delta\text{BloU} = -0.1194$) despite the modest Dice change.

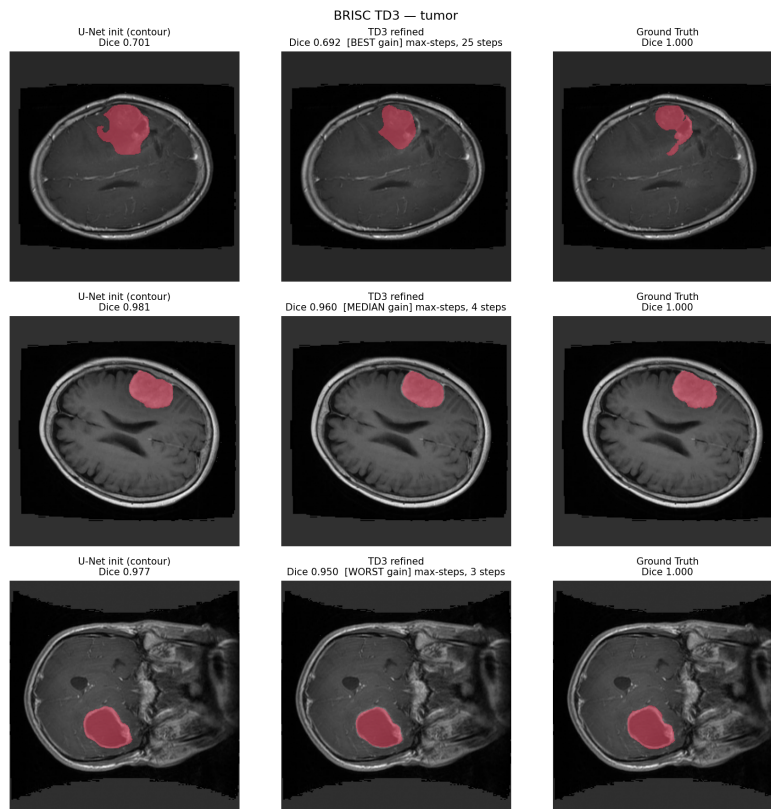


Figure 5.4 — BRISC tumor, TD3, Phase A (full data). Deployed $\Delta\text{Dice} = -0.0023$, the smallest BRISC degradation of the four BRISC runs — consistent with TD3 being the safer agent specifically on this dataset.

6. High-Data (Phase A) vs. Low-Data (Phase B)

Phase	Mean Δ Deployed Dice	Mean Δ IoU	Mean Δ BIoU	Mean HD95 (init $\rightarrow$ final)
Phase A (full)	-0.0071	-0.0326	-0.0577	6.60 $\rightarrow$ 7.53
Phase B (low)	-0.0028	-0.0182	-0.0253	10.95 $\rightarrow$ 11.54

The average degradation is smaller under low data than under full data — Phase B's mean deployed-Dice change (-0.0028) is roughly a third the size of Phase A's (-0.0071), and the same pattern holds for IoU and Boundary IoU.

This is NOT evidence that DRL's relative advantage grows under scarcity

That was the hypothesis the project's own literature review set out to test. Both regimes are negative; neither shows DRL actually improving on the baseline. What this data actually shows is narrower: DRL does *less harm*, on average, when the baseline it refines is already weaker (Phase B's AttentionResUNet baseline Dice is itself notably lower than Phase A's — e.g. BRISC 0.7555 vs. 0.8766). A plausible mechanism is that the value-floor has more headroom to matter when the starting point is already imperfect. This is a hypothesis for the paper to state carefully, not a confirmed causal finding — the sample size here (4 classes $\times$ 2 agents per phase) is too small for a real significance test on this specific comparison.

Win/tie/loss by phase (same ± 0.005 band): DuelingDDQN is 3-tie/1-loss under full data and a clean 4-tie/0-loss under low data. TD3 is 1-tie/3-loss under full data and 3-tie/1-loss under low data. Both agents' loss count drops under low data, reinforcing the pattern above.

7. Does DRL Actually Refine the Mask?

7.1 Quantitatively: no

Sections 4–6 already establish this; it bears restating as the single clearest sentence for the paper: **none of the 16 real runs improved deployed Dice beyond noise, and boundary-precision metrics (Boundary IoU, HD95) got measurably worse on average in every phase/agent combination** — including cases where Dice itself looked close to flat. BRISC/tumor DuelingDDQN under full data is the clearest example: Dice moved only -0.0094, but Boundary IoU dropped -0.1194 and HD95 grew from 6.60px to 9.09px.

A near-tied Dice number can hide a real loss of boundary quality

The paper should not report Dice alone as "no change" without also reporting Boundary IoU and HD95 for the same run — this is one of the more defensible and interesting findings in this dataset, and it is easy to miss if only Dice is reported.

7.2 Qualitatively: by eye, sometimes yes on shape

The exported comparison-grid figures (Section 5.1 above, and the additional examples below) show, by eye, refined masks that are visually recognisable approximations of the true anatomical or tumour shape in most of the best- and median-gain rows — smoother, more contour-like boundaries than a raw pixel mask might show, generally the right size, location, and shape for the target structure. The worst-gain rows show visibly larger deviations — boundary drift, partial coverage, or shape distortion — consistent with the quantitative degradation reported above.

This qualitative read needs real hedging, not dressing up as a second quantitative result

It comes from a visual read of a handful of exported comparison grids, not a blinded, structured, or reader-scored assessment. It is legitimate to write in a paper as "informal visual inspection suggests the refined contours remain anatomically plausible even where they do not improve on standard overlap metrics" — but it should not be presented as a rigorous qualitative evaluation. See Section 9, Recommendation 5, for what a real qualitative study would need.

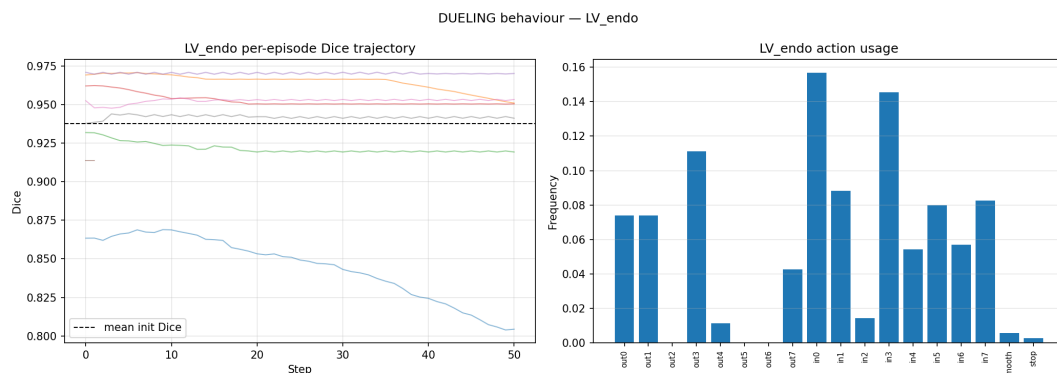


Figure 7.1 — CAMUS LV_endo, DuelingDDQN, Phase A: per-episode Dice trajectory (left) and action-usage histogram (right) across the sampled test episodes.

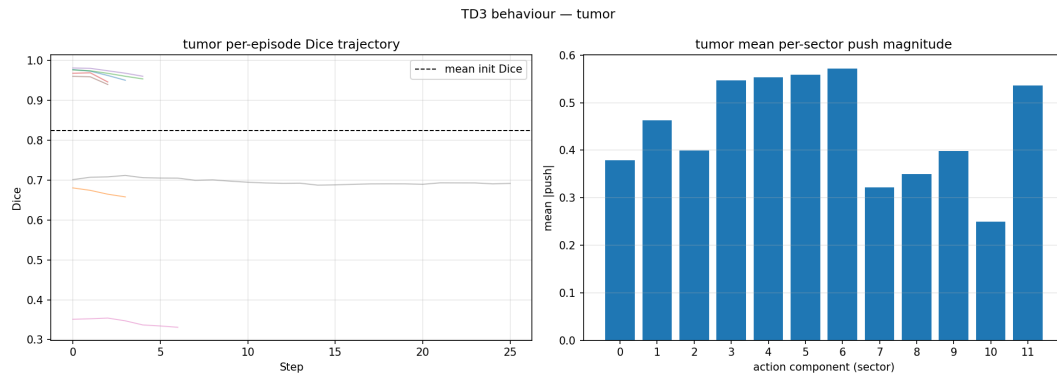


Figure 7.2 — BRISC tumor, TD3, Phase A: per-episode Dice trajectory (left) and mean per-sector push magnitude (right). Compare the trajectory shape against Figure 7.1 — TD3's continuous per-step magnitude gives a visibly different exploration pattern than DuelingDDQN's discrete pushes.

7.3 Future-proofing for a real paired qualitative study

The evaluation notebook's Section 9a already groups every model's and agent's qualitative images by (dataset, class, phase) so that all of a target's outputs are reviewed together. A patch to `iteris/refinement_viz.py` (commit 9ceea97) now also labels each future comparison-plot row with its patient ID — a field that already existed on every replay's sample but was not previously drawn. A **future** Kaggle re-run would therefore support matching the exact same patient across models and agents for a genuinely paired qualitative comparison, which the current export does not support (each agent's best/median/worst rows are its own independently-picked patients).

8. What This Report Deliberately Does Not Claim

Per the explicit instruction to avoid overstated or misleading comparisons:

- **No LiteUNet-vs-DRL comparison anywhere in this report.** LiteUNet is not the DRL target (single-baseline design), and comparing e.g. a Phase B LiteUNet number against a Phase A DRL number would combine two unrelated variables — weaker architecture *and* less data — into one misleading number in either direction. LiteUNet's only legitimate role in this project is the separate architecture-headroom comparison in the evaluation notebook's Section 3, which this report does not repeat.
- **No claim about the BRISC tumor-type classifier** — it was not part of this data export.
- **No formal significance test (Wilcoxon signed-rank / Bonferroni).** That requires per-patient paired data (a `*_test_scores.csv` for U-Net, or a per-sample DRL export), neither of which was attached to this run. The 16-run table in Section 4.1 is a census of every available run, not a statistical sample — with $n=4$ classes per phase/agent cell it is too small to support a hypothesis test with real power, and the paper should not present p-values computed on it.
- **No claim that Phase B "helps DRL more."** See Section 6's explicit caveat — less harm is not more help.

9. Recommendations for the Capstone Write-Up

- 1 **Lead with the honest quantitative finding** (Section 4): the value floor makes DRL safe — mostly ties, never a clear loss beyond the BRISC/LV_epi-TD3 outliers — but not yet beneficial. Frame the contribution as "a safe, deployable contour-refinement mechanism that does not yet close a measurable Dice gap," rather than overselling a positive result that is not in this data.
- 2 **Report Boundary IoU and HD95 alongside Dice for every headline number.** Section 7's finding that boundary quality can degrade even when Dice looks flat is one of the more interesting, defensible things in this dataset and is easy to miss if only Dice is reported.
- 3 **Discrete vs. continuous:** present DuelingDDQN as the safer default, but discuss BRISC as a genuine counter-example (small-target volatility, Section 5) rather than presenting one agent as universally better.
- 4 **Before writing the Phase A/B section,** get the Kaggle export folder-naming fixed (Section 3) so future runs do not need this correction re-derived by hand, and ideally re-attach a run with `label_frac` correctly present on the U-Net baseline JSONs so the "Unknown"-phase caveat in Section 3.3 goes away too.
- 5 **If a formal qualitative section is wanted for the paper,** budget for an actual structured rating pass (even 2–3 independent raters scoring best/median/worst samples) rather than relying on the informal read in Section 7.2 — the infrastructure for it (Section 9a's grouped gallery, the patient-ID-labelled future exports) is already in place and just needs a fresh Kaggle run plus a rating protocol.

Appendix A: Full Per-Run Metric Detail

Every field below is transcribed directly from the exported run JSONs (*_test_results.json), with the phase correction from Section 3 already applied. All Dice/IoU/BloU values are dimensionless [0,1]; HD95 and MSD are in pixels.

Class	Agent	Ph	Init Dice	Final Dice	Deploy Dice	Init HD95	Final HD95	Init IoU	Final IoU	Init MSD	Final MSD
CAMUS/LV_endo	DuelingDDQN	A	0.9356	0.9301	0.9338	5.71	6.05	0.8815	0.8720	2.15	2.30
CAMUS/LV_endo	TD3	A	0.9356	0.9205	0.9241	5.71	6.55	0.8815	0.8550	2.15	2.66
CAMUS/LV_endo	DuelingDDQN	B	0.9129	0.9106	0.9116	8.01	8.19	0.8431	0.8393	2.92	3.01
CAMUS/LV_endo	TD3	B	0.9129	0.9068	0.9128	8.01	8.27	0.8431	0.8325	2.92	3.15
CAMUS/LV_epi	DuelingDDQN	A	0.8621	0.8592	0.8612	6.59	6.74	0.7607	0.7562	2.52	2.57
CAMUS/LV_epi	TD3	A	0.8621	0.8354	0.8420	6.59	8.58	0.7607	0.7199	2.52	3.08
CAMUS/LV_epi	DuelingDDQN	B	0.8166	0.8157	0.8163	10.02	10.06	0.6952	0.6938	3.54	3.56
CAMUS/LV_epi	TD3	B	0.8166	0.7891	0.8022	10.02	12.36	0.6952	0.6558	3.54	4.22
CAMUS/LA	DuelingDDQN	A	0.8962	0.8859	0.8937	7.48	7.96	0.8220	0.8058	2.79	3.03
CAMUS/LA	TD3	A	0.8962	0.8777	0.8883	7.48	8.31	0.8220	0.7914	2.79	3.26
CAMUS/LA	DuelingDDQN	B	0.8526	0.8398	0.8518	11.26	12.40	0.7582	0.7399	4.18	4.54
CAMUS/LA	TD3	B	0.8526	0.8315	0.8515	11.26	11.71	0.7582	0.7258	4.18	4.65
BRISC/tumor	DuelingDDQN	A	0.8766	0.8177	0.8672	6.60	9.09	0.8115	0.7193	2.43	3.12
BRISC/tumor	TD3	A	0.8766	0.8523	0.8743	6.60	6.95	0.8115	0.7710	2.43	2.78
BRISC/tumor	DuelingDDQN	B	0.7555	0.7475	0.7526	14.50	14.73	0.6783	0.6687	7.17	7.31
BRISC/tumor	TD3	B	0.7555	0.7368	0.7540	14.50	14.56	0.6783	0.6484	7.17	7.45

Appendix B: Glossary

Dice — Overlap-based segmentation accuracy metric; $2 \times |A \cap B| / (|A| + |B|)$, range $[0,1]$, higher is better.

IoU — Intersection-over-Union; $|A \cap B| / |A \cup B|$, range $[0,1]$, higher is better.

Boundary IoU (BIOU) — IoU computed only within a narrow band around the mask boundary — sensitive to boundary precision in a way whole-mask Dice/IoU are not.

HD95 — 95th-percentile Hausdorff Distance in pixels — a boundary-distance error metric; lower is better.

MSD — Mean Surface Distance in pixels; lower is better.

Init Dice / Init HD95 / etc. — The metric value of the AttentionResUNet baseline mask, before any DRL refinement, on the env's own rasterised contour representation.

Final Dice — The raw Dice of the mask at the end of the DRL episode (max-steps or a deliberate STOP), with no safety floor applied.

Deployed / value-floored Dice — The Dice of whichever state the agent's own value estimate rated highest across the episode, falling back to the initial mask if nothing scored higher — the number that would actually ship.

Value floor / Pillar 1 — The GT-free "do no harm" deployment rule described in Section 1.4.

DuelingDDQN — Dueling Double Deep Q-Network; the discrete agent, 18 actions (8 sectors $\times$ push out/in, plus smooth and STOP).

TD3 — Twin Delayed Deep Deterministic Policy Gradient; the continuous agent, per-sector signed displacement.

Phase A / Phase B — The full-data and low-data (≈ 150 -sample) experimental regimes, respectively — see Section 3 for the label-inversion correction applied throughout this report.

AttentionResUNet — The strong U-Net architecture both agents refine directly (the single-baseline design).

LiteUNet — A separate, deliberately lighter U-Net architecture kept only as an architecture-comparison reference; not refined by either agent.